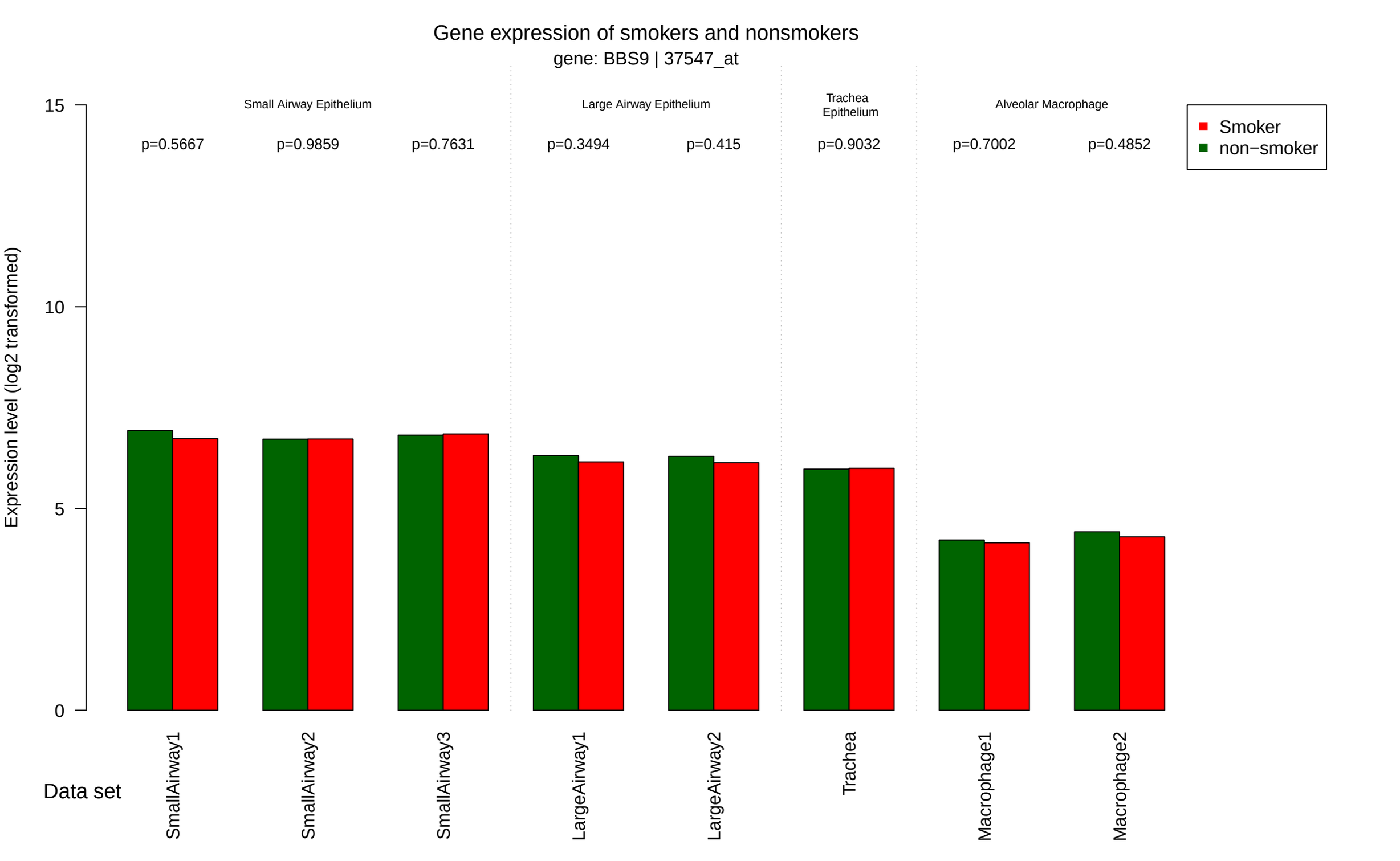
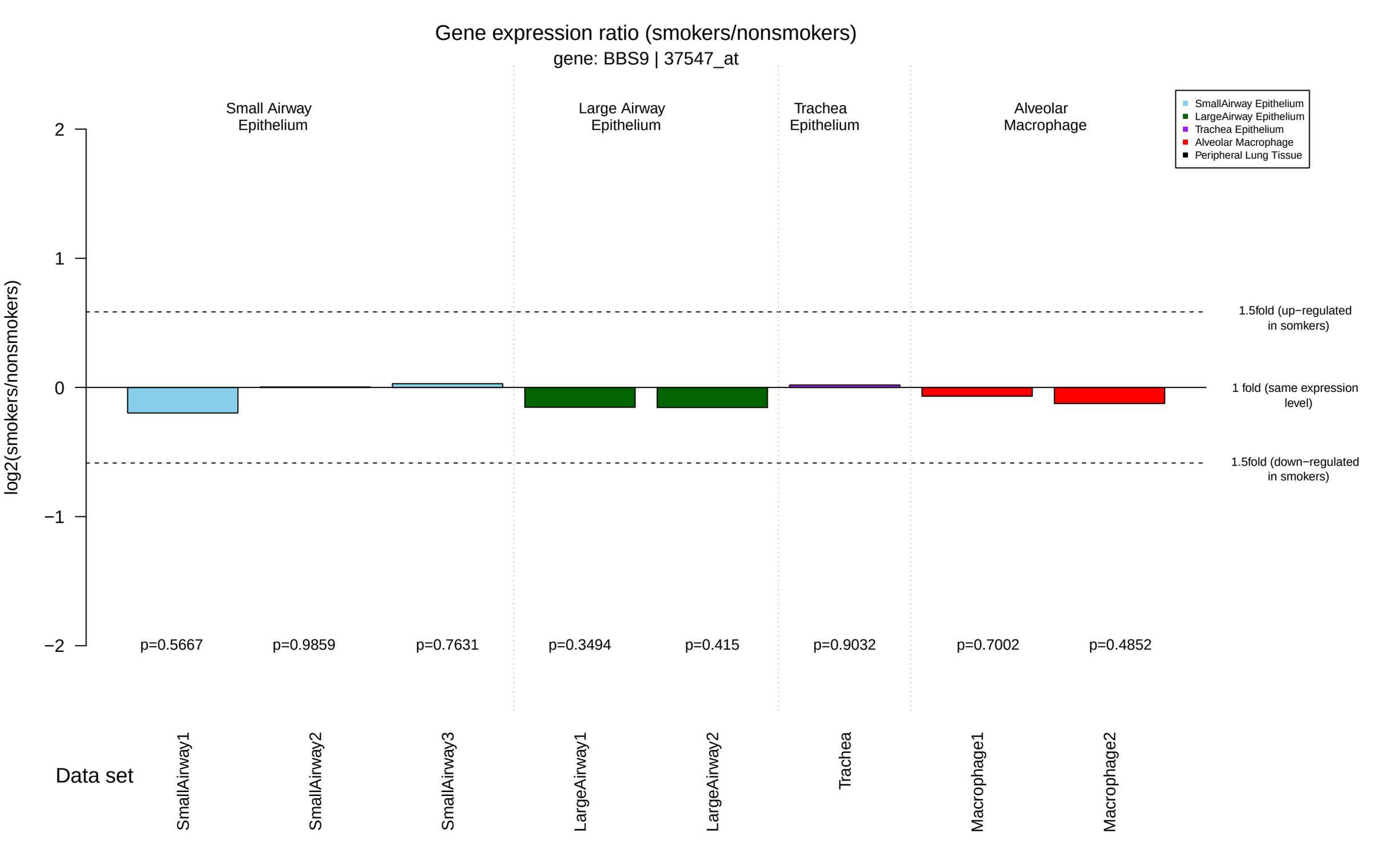
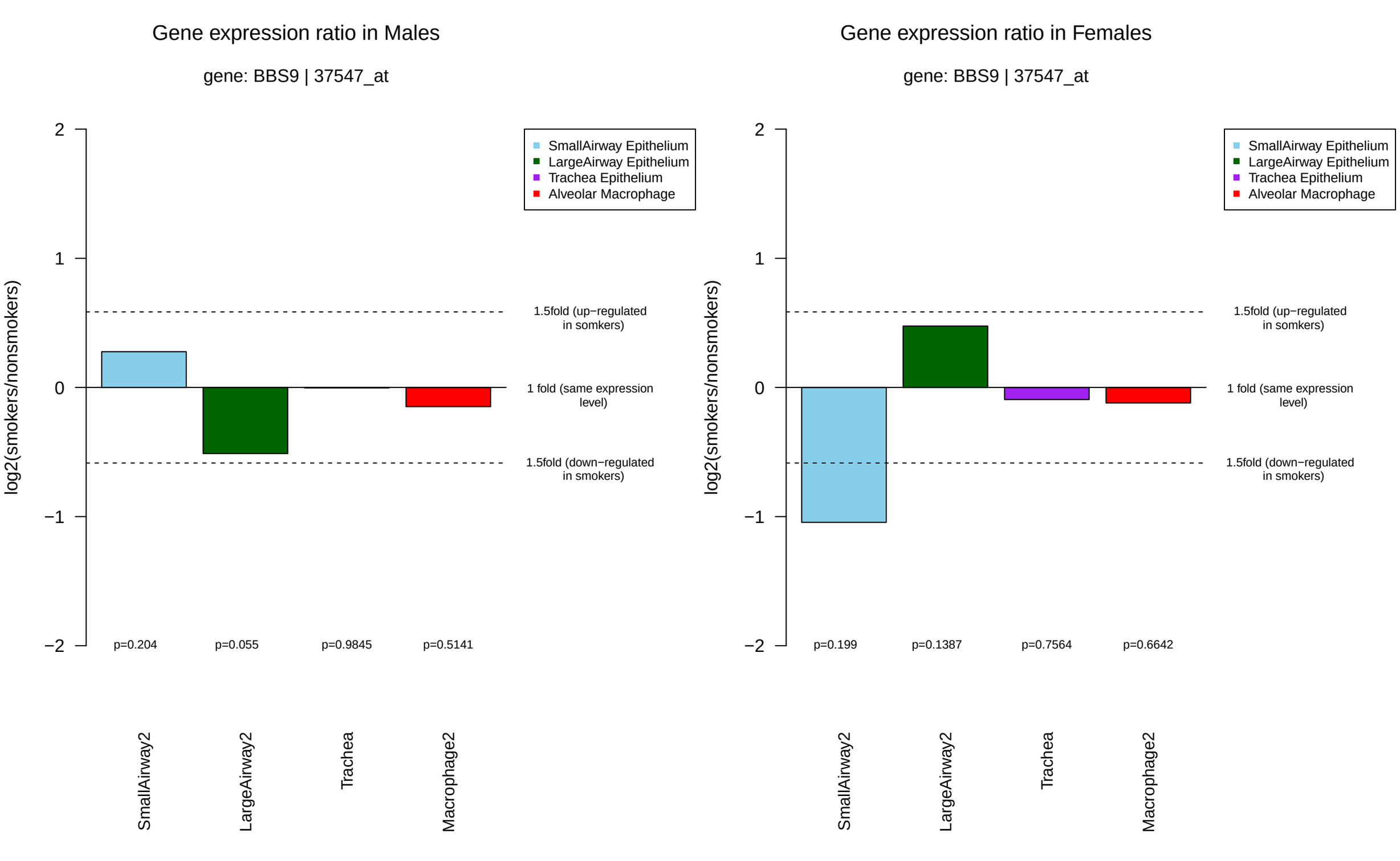
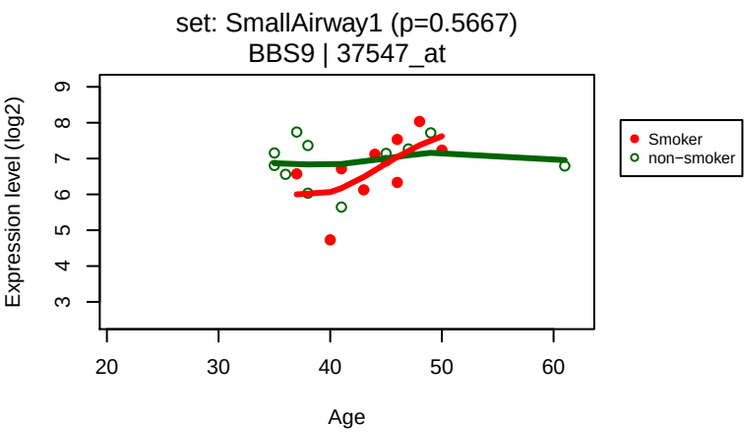
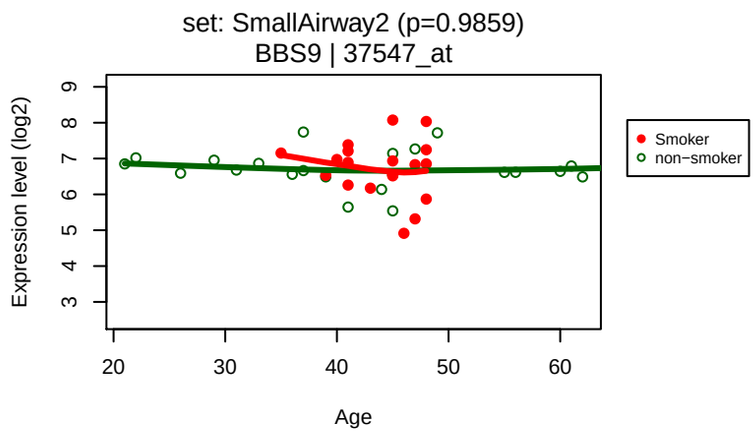
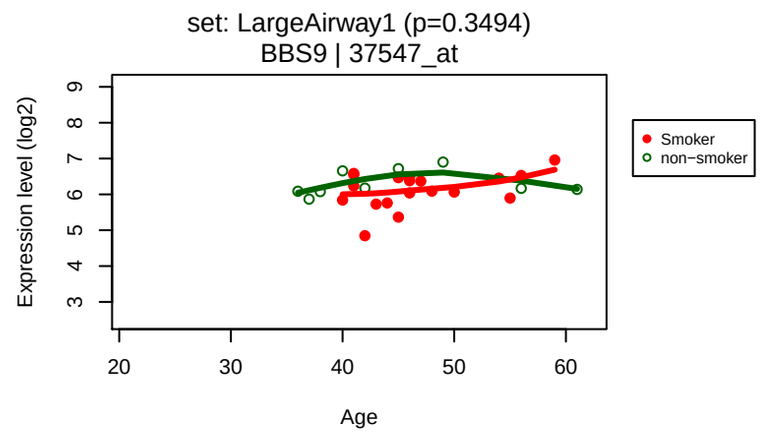

Gene expression of smokers and nonsmokers
gene: BBS9 | 37547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5667
p=0.9859
p=0.7631
p=0.3494
p=0.415
p=0.9032
p=0.7002
p=0.4852
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

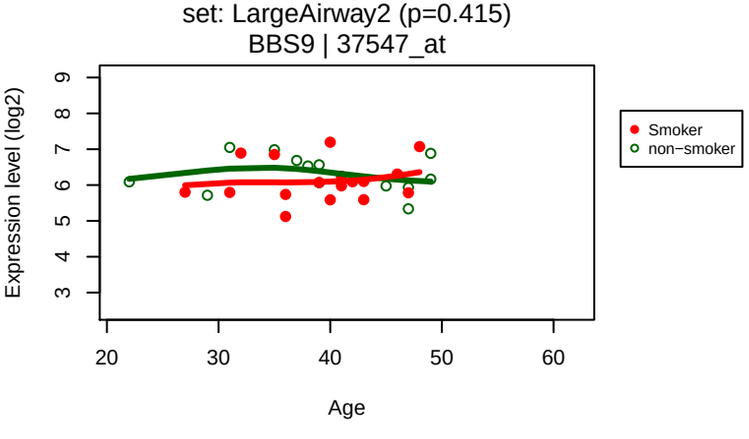

Gene expression ratio (smokers/nonsmokers)
gene: BBS9 | 37547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5667
p=0.9859
p=0.7631
p=0.3494
p=0.415
p=0.9032
p=0.7002
p=0.4852
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

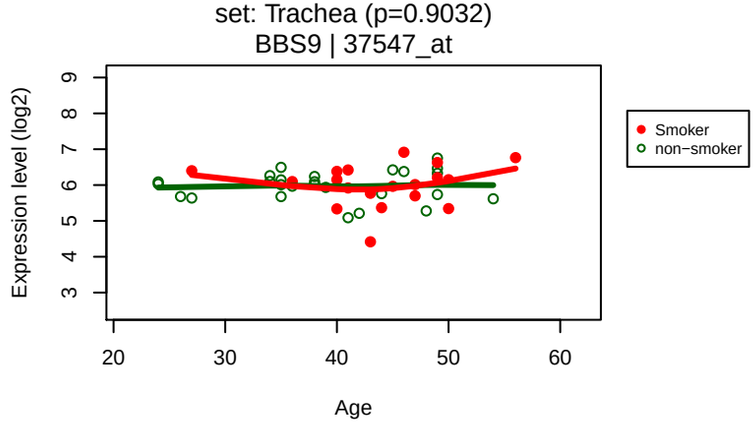

Gene expression ratio in Males
gene: BBS9 | 37547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.204
p=0.055
p=0.9845
p=0.5141
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BBS9 | 37547_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.199
p=0.1387
p=0.7564
p=0.6642
SmallAirway2
LargeAirway2
Trachea
Macrophage2

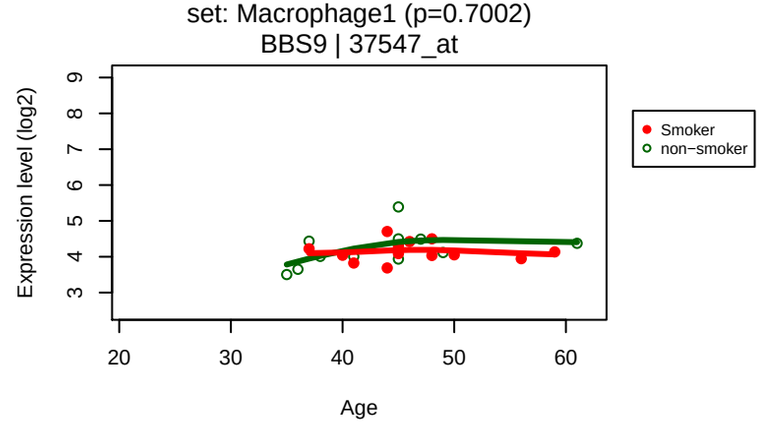

set: SmallAirway1 (p=0.5667)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker

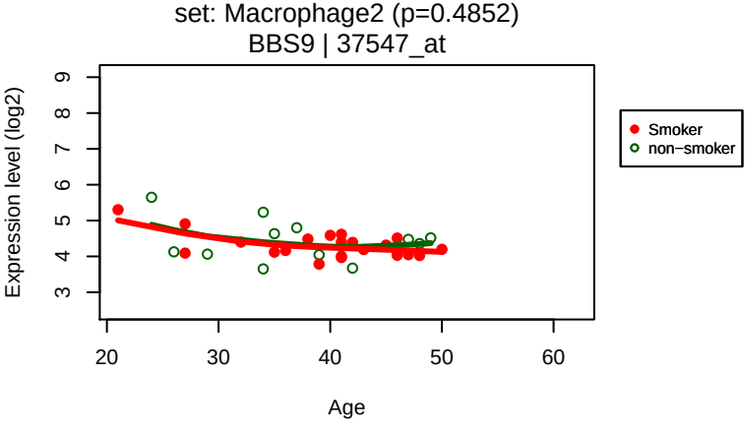

set: SmallAirway2 (p=0.9859)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker

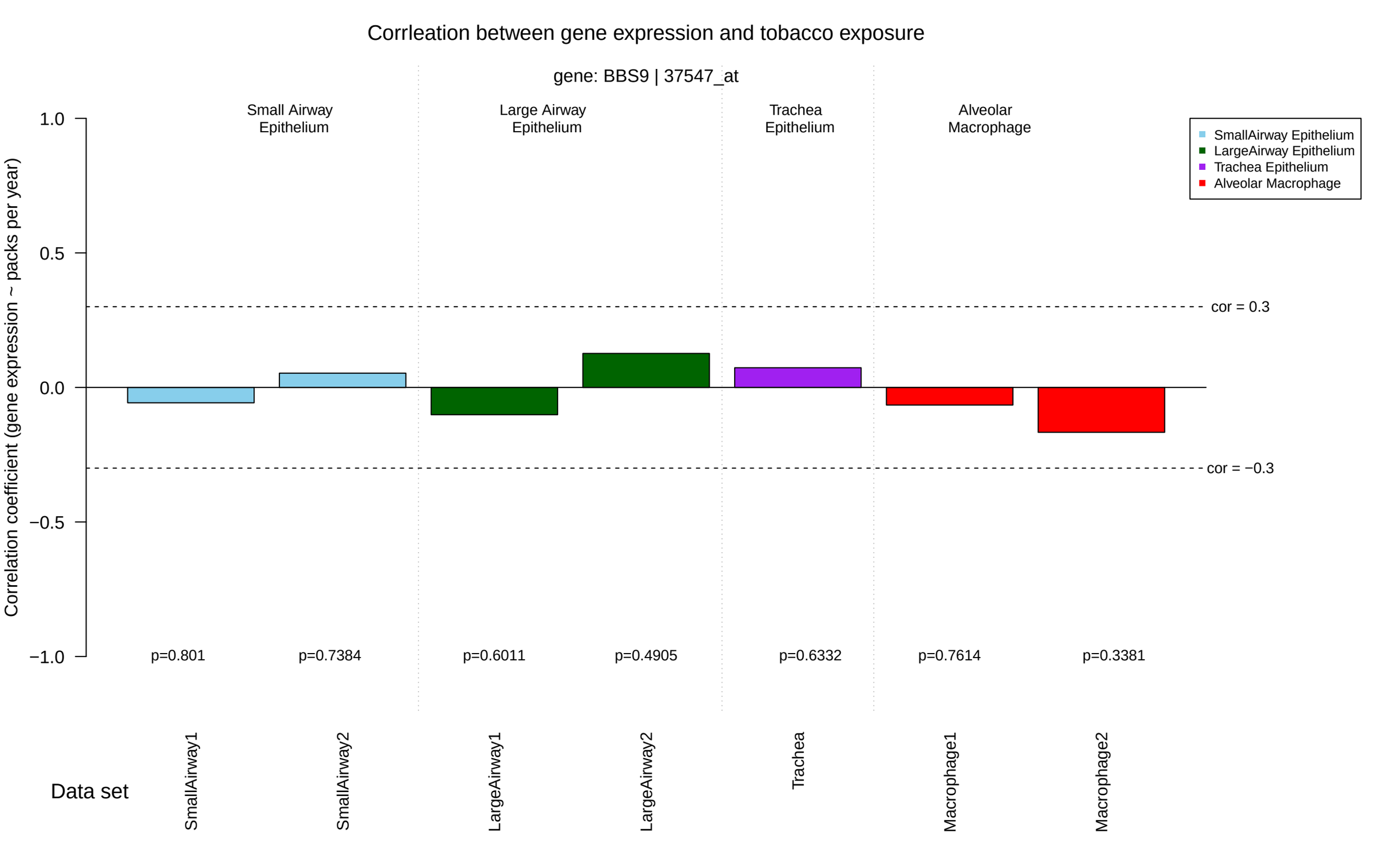

set: LargeAirway1 (p=0.3494)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.415)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9032)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7002)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4852)
BBS9 | 37547_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BBS9 | 37547_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.801
p=0.7384
p=0.6011
p=0.4905
p=0.6332
p=0.7614
p=0.3381
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2